

FINAL EXAMINATION ANSWERS

- (1) State the two Fundamental Theorems of Calculus. Give an example (with proof) of a function $f : [0, 1] \rightarrow \mathbb{R}$ which is not integrable but f^2 is integrable. (6+8 = 14 marks)

Answer: The statements of the two Fundamental theorems of Calculus are as follows.

Theorem I: If g is a continuous real valued function on $[a, b]$ which is differentiable on (a, b) and if g' is integrable on $[a, b]$, then $\int_a^b g' = g(b) - g(a)$.

Theorem II: Let f be an integrable function on $[a, b]$. For x in $[a, b]$, let $F(x) = \int_a^x f(t)dt$. Then F is continuous on $[a, b]$. If f is continuous at x_0 in (a, b) , then F is differentiable at x_0 and $F'(x_0) = f(x_0)$.

An example of a function f on $[0, 1]$ which is not integrable, but f^2 is integrable, is given by $f(x) = 1$ if x is rational and $f(x) = -1$ if x is irrational. The proof that f is not integrable should be done as in Example 2, page 272 of the book by Ross. Note the function in Ross Example 2 of page 272 is slightly different, the two values f takes there are 0 and 1. In our case $U(f, P) = 1$ and $L(f, P) = -1$ for any partition of $[0, 1]$.

- (2) Let $A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix}$. Consider the system of linear equations $Ax = b$ where $x = (x_1, x_2, x_3)^T$ is the column of three variables, and $b = (5, -2, 9)^T$. Solve this system of equations using Gaussian elimination. Write down the LU decomposition of A . Use the Gauss-Jordan method to find the inverse of the matrix A . (4+4+4=12 marks)

Answer: This problem is done in Strang in full detail. See Section 1.3 page 13 for the first part (solve using Gaussian elimination), Section 1.5 pages 37 and 38 for the second part (write the LU decomposition) and Section 1.6 page 53 for the third part (find inverse using Gauss-Jordan method). The above system of linear equations, using Gaussian elimination, leads to the equivalent system

$Ux = c$ where $U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{pmatrix}$ and $c = (5, -12, 2)^T$. This gives via back substitution, $x_3 = 2$, $x_2 = 1$ and $x_1 = 1$. The LU decomposition of A is given

by $U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{pmatrix}$ and $L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}$. We start with the follow-

ing matrix and use the Gauss-Jordan method for finding the inverse as follows

$\begin{pmatrix} 2 & 1 & 1 & 1 & 0 & 0 \\ 4 & -6 & 0 & 0 & 1 & 0 \\ -2 & 7 & 2 & 0 & 0 & 1 \end{pmatrix}$. This leads to the following matrices (step by step using row operations) $\begin{pmatrix} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & -8 & -2 & -2 & 1 & 0 \\ 0 & 8 & 3 & 1 & 0 & 1 \end{pmatrix}$, and then $\begin{pmatrix} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & -8 & -2 & -2 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{pmatrix}$, next $\begin{pmatrix} 2 & 1 & 0 & 2 & -1 & -1 \\ 0 & -8 & 0 & -4 & 3 & 2 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{pmatrix}$, and then $\begin{pmatrix} 2 & 0 & 0 & \frac{12}{8} & -\frac{5}{8} & -\frac{6}{8} \\ 0 & -8 & 0 & -4 & 3 & 2 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{pmatrix}$, and finally $\begin{pmatrix} 1 & 0 & 0 & 12/16 & -5/16 & -6/16 \\ 0 & 1 & 0 & 4/8 & -3/8 & -2/8 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{pmatrix}$. So A^{-1} is the matrix $\begin{pmatrix} 12/16 & -5/16 & -6/16 \\ 4/8 & -3/8 & -2/8 \\ -1 & 1 & 1 \end{pmatrix}$, which is the same as $\begin{pmatrix} 3/4 & -5/16 & -3/8 \\ 1/2 & -3/8 & -1/4 \\ -1 & 1 & 1 \end{pmatrix}$.

(3) Let $A = \begin{pmatrix} 1 & 2 & 3 & 5 \\ 2 & 4 & 8 & 12 \\ 3 & 6 & 7 & 13 \end{pmatrix}$. Consider the system of linear equations $Ax = b$ where

$x = (x_1, x_2, x_3, x_4)^T$ is the column of the four variables, and $b = (b_1, b_2, b_3)^T \in \mathbb{R}^3$. Use elimination to reduce $Ax = b$ to $Ux = c$ and $Rx = d$, where U and R are the echelon and the reduced echelon forms respectively. Find the condition on b_1, b_2, b_3 for a solution of $Ax = b$ to exist. Find a particular solution to $Ax = b$, if $b = (0, 6, -6)^T$. Find all solutions to the system $Ax = (0, 6, -6)^T$ by finding all solutions of the homogeneous system $Ax = 0$. (3+3+3+3=12 marks).

Answer: This problem is done in Strang in full detail, see pages 94 and 95, Chapter 2, Section 2 of Strang for the details. We use elimination to reduce $Ax = b$ to $Ux = c$ as follows. We have

$\begin{pmatrix} 1 & 2 & 3 & 5 & b_1 \\ 2 & 4 & 8 & 12 & b_2 \\ 3 & 6 & 7 & 13 & b_3 \end{pmatrix}$, this gives $\begin{pmatrix} 1 & 2 & 3 & 5 & b_1 \\ 0 & 0 & 2 & 2 & b_2 - 2b_1 \\ 0 & 0 & -2 & -2 & b_3 - 3b_1 \end{pmatrix}$. This in turn gives $\begin{pmatrix} 1 & 2 & 3 & 5 & b_1 \\ 0 & 0 & 2 & 2 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 + b_2 - 5b_1 \end{pmatrix}$. This

is the $Ux = c$ echelon form. It also gives the condition on b_1, b_2, b_3 for a solution of $Ax = b$ to exist, the condition is $b_3 + b_2 - 5b_1 = 0$. Next we get the $Rx = d$

form which is $\begin{pmatrix} 1 & 2 & 0 & 2 & 4b_1 - 3b_2/2 \\ 0 & 0 & 1 & 1 & b_2/2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 + b_2 - 5b_1 \end{pmatrix}$. If $b = (0, 6, -6)^T$, then a particular

solution to $Ax = b$ is a particular solution to both $Ux = c$ and $Rx = d$, where $c = (0, 6, 0)^T$ and $d = (-9, 3, 0)^T$. It is clear that x_1 and x_3 are pivot variables and x_2 and x_4 are free variables and so we get a particular solution to $Rx = d$ by putting $x_2 = x_4 = 0$, hence a particular solution is $(-9, 0, 3, 0)^T$. The solutions to the homogeneous system $Ax = 0$ are the same as the solutions to the homogeneous systems $Ux = 0$ and $Rx = 0$ and any solution of $Rx = 0$ is a linear combination of two solutions which are obtained by putting $x_2 = 1, x_4 = 0$ and $x_2 = 0, x_4 = 1$

respectively. These two solutions are $(-2, 1, 0, 0)^T$ and $(-2, 0, -1, 1)^T$. So any solution of $Ax = b$ is of the form $(-9, 0, 3, 0)^T + c_1(-2, 1, 0, 0)^T + c_2(-2, 0, -1, 1)^T$ where c_1 and c_2 are any two real numbers.

- (4) Find the dimension and a basis for each of the four fundamental vector spaces associated to the matrix $A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix}$. The four vector spaces are the column space $C(A)$, the null space $N(A)$, the row space $C(A^T)$ and the left null space $N(A^T)$. Here A^T denotes the transpose of the matrix A . (3+3+3+3=12 marks)

Answer: It is clear that the reduced echelon form is $R = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. So

$m = 3$, $n = 4$, $r = 2$. Thus both $C(A)$ and $C(A^T)$ has dimension 2, $N(A)$ has dimension 2 and $N(A^T)$ has dimension 1. Clearly the pivot columns $(1, 0, 1)$ and $(0, 1, 0)$ are a basis of $C(A)$ and the pivot rows $(1, 2, 0, 1)$ and $(0, 1, 1, 0)$ form a basis for $C(A^T)$. A basis for $N(A^T)$ is the vector $(1, 0, -1)$ since the first and third rows of A are the same. Finally the pivot variables are x_1 and x_3 and the free variables are x_2 and x_4 and from the reduced echelon form R , the two equations of the homogeneous system are $x_1 + 2x_2 + x_4 = 0$ and $x_2 + x_3 = 0$. If we put $x_2 = 1$ and $x_4 = 0$, we get the solution $(-2, 1, -1, 0)$. If we put $x_2 = 0$ and $x_4 = 1$, we get the solution $(-1, 0, 0, 1)$. So a basis of $N(A)$ are the two vectors $(-2, 1, -1, 0)$ and $(-1, 0, 0, 1)$.